

Ethernet and Fast Ethernet

Module 23

Duration: 29:01

Video: <https://www.youtube.com/watch?v=AN-s1J2GrPk>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Prof Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

Auto-generated YouTube transcript for personal study. Spoken wording belongs to the original lecture. Follow the video link above; this file is not an official CEC/SWAYAM publication.

Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar University of Science and Technology Hisar harana I welcome you all to the video lecture series of cyber security and information secur in this lecture we will study about ethernet and fast ethernet which is basically used for data transmission on wired Lan at a speed of 10 and 100 Mbps respectively first of all please see the contents which I shall cover after covering brief introduction about ethernet I shall take up protocol architecture and frame format of ethernet which will be followed by Mac operations in the ethernet then I shall take up fast ethernet and protocol architecture of fast

ethernet so now let us start ethernet is a family of computer networking Technologies for local area network and metropolitan area network it was developed as an experimental coxl cable network in 1970s by jerox Corporation initially it operated with a data rate of just 3 Mbps using a carrier sense multiple assess with Collision detection protocol for Lan it was commercially introduced in 1980 and standardized in 1983 as i e 8 2 minutes, 1 second 2.3 it has since been refined to support higher bit rates and longer link distances over time ethernet has largely replaced competing wired land Technologies such as token ring token bus fddi and arknet ethernet land topology is currently the most common Network architecture

ethernet topologies are generally bus or bus star topologies the I 802.3 standard defines ethernet protocols for OSI media access control sublayer and physical layer Network characteristics the name ethernet refers to the cable ether four types of cablingserver ethernet as shown in table and figure table shows four types of cable standards however figure part A shows 10 base five connections Part B shows 10 base 2 connections and part C shows 10 base T connections historically 10 base 5 cabling popularly called thick ethernet came first connections to it are generally made using vampire tabs as shown in part A of figure in which a pin is very carefully forced halfway into the coxial caes core in Ethernet standard the first number is the speed in

Mbps then comes the word base to indicate base band transmission 4 minutes, 1 second accordingly the notation 10 base 5 means that it operates at 10 Mbps using baseband signals and can support segments of up to 500 M the second cable type 10 base 2 was made up of thin coaxial cable using BNC connectors to form P Junctions rather than using vampire tabs as shown in Part B of figure thin ethernet is much cheaper and easier to install but it can run for only 185 m per segment Each of which can handle only 30 machines problem with 10 base 5 and 10 Bas 2 is that whole network runs on single Cable in case cable gets fault at some place whole network goes down the problems associated with finding cable brakes drove systems toward a different kind of wiring

pattern in which all stations have a cable running to a central Hub this scheme is called 10 Bas T in this scheme each computer is connected to HUB through separate twisted pair cable as shown in part C of figure it supports 100 m distance with 1024 nodes and is cheapest system a fourth cabling option for ethernet is 10 Bas F which uses fiber Optics this 6 minutes, 1 second alternative is expensive due to the cost of the connectors and Terminators but it has excellent noise immunity and is the method of choice when running between buildings or widely separated hubs runs of up to 2,000 M are allowed it also offers good security since wire tapping a fiber is much more difficult than wire tapping a copper wire it supports 1024 nodes and is best

suitable for connecting two buildings now let us see ethernet layered protocol architecture in Ethernet both the data link and the physical layers are involved in the creation and transmission of frames the physical layer is related to the type of Lan cabling and how the bits are transmitted and received on the cable the data link layer is divided into Su layers which are logical link control LLC and media access control Mac the LLC sublayer is responsible for identifying and passing data to the network layer protocol and Max sub layer is responsible for medium access control for shared data the MAC sublayer address is the physical Hardware address of the source and destination computer this sublayer controls which

computer devices send and receive the data and allows ni is to communicate with the physical layer the frame format of i e 802.3 is as shown in figure the Preamble of the frame which is of 7 bytes indicates the start of a new frame and establishes synchronization conditions between devices the last bite of Preamble known as start of frame delimiter soff always has a 10101010 bit pattern this bite indicates the start of a frame the destination address of 6 bytes is the hardware Mac address of the receiving device and the source address of 6 bytes specifies the hardware Mac address of of the sending device the length field of 2 bytes indicates the length of frame the data field is for the actual data being

transmitted from device to device and length of data field is 46 to 1500 bytes in case length of data is less than 46 bytes padding is done to make it 46 byte the cyclic redundancy check CRC of four bytes checks that the frame is received free from error after frame format let us see how medium access control works in Ethernet protocols in which stations listen for a carrier

and act accordingly are called carrier sense protocol s ethernet uses CSM by CD as medium assess control protocol for channel sensing when a station has data to send it first listens to the channel to see if anyone else is transmitting at that moment if the channel is busy the station Waits until it becomes idle when the station detects an idle Channel it

transmits a frame if a collision occurs the station Waits a random amount of time and starts all over again the process of csma by CD is shown in flowchart in this flowchart when a station is ready to send data it senses the channel if channel is busy it makes a new attempt to set s the channel if 11 minutes, 1 second channel is free it transmits data and senses the channel for Collision in case collisions are detected it transmits a jam signal to indicate that Collision has occurred and it wait as per back of strategy if no collisions are detected transmission completes CSM by CD uses the conceptual model shown in figure CSM by CD can be in one of three states contention transmission or idle at the point marked t_0 a station has finished transmitting its

frame any other station having a frame to send may now attempt to do so if two 12 minutes, 1 second or more stations decide to transmit simultaneously there will be a collision collisions can be detected by looking at the power or pulse width of the received signal and comparing it to the transmitted signal after a station detects a collision it aborts its transmission weits a r period of time and then tries again assuming that no other station has started transmitting in the meantime therefore our model for csma by CD will consist of alternating contention and transmission periods with ideal periods occurring when all stations are quiet now let us see what happens if Collision occurs after

the first Collision each station Waits either zero or one slot times before trying again if two stations Collide and each one picks the same random number they will collide again after the second Collision each one picks either 0o 1 2 or three at random and waits that number of slot times if a third Collision occurs then again the number of slots to wait is chosen at random from the interval 0 to $2^3 - 1$ that is 7 in journal after I collisions a random number between zero and $2^I - 1$ is chosen and that number of slots is skipped however after 10 collisions have been reached the randomization interval is Frozen at maximum of 1023 slots after 16 collisions the controller throws in the towel and

reports failure back to the computer further recovery is up to the higher layers this algorithm is called binary exponential back off now let us see the reason for having padding bits in Ethernet frame reason for having a minimum length frame is to prevent a station from completing the transmission of a short frame before the first bit has even reached the far end of the cable where it may collide with another frame this problem is Illustrated in figure in figure a at Time Zero station a at one end of the network sends off a frame let T be the propagation time for this Frame to reach the other end B in figure B just before the frame gets to the other end that is at time T minus E the most distant station B starts transmitting as shown in figure C when B

detects that it is receiving more power than it is putting out it knows that a collision has occurred so it aborts its transmission and generates a 40 8bit noise burst to warn all other stations in other words it jams The Ether to make sure the sender does not miss the collision at about time $2T$ the sender sees the noise burst and about its transmission as shown in figure D it then Waits a random time before trying again if a station tries to transmit a very short frame and collision occurs the sender will then incorrectly conclude that the frame was successfully sent to prevent this situation from occurring all frames must take more than $2T$ time to send data frame so that the transmission is still taking place when

and the noise burst gets back to the sender that is the reason why padding bits are added so that transmission is not complete in $2T$ time frames with fewer than 64 bytes are padded out to 64 bytes with a pad field so this was about ethernet now let us move on to fast ethernet fast ethernet is traditional csma by CD at 100 Mbps over twisted pair cable the idea of fast ethernet was first proposed in 1992 during the early development of fast ethernet two different groups worked out standard proposals and both were finally approved but under different itle committees one standard passed a full review in June 1995 and was formally assigned the name 802.3 U and the other became 100 VG any Lan which is now governed by I 802.2

committee the letter uses the demand priority media exess method instead of CSM by CD method 100 VG any Lan has not caught on while fast ethernet enjoyed great success in Enterprise Lan environment due to slow speed of ethernet fast ethernet came into existence which is simple speed up version of ethernet with same rules and regulations the 802.3 committee decided to go with a speed up ethernet for three primary reasons one the need to be backward compatible with existing ethernet Lens two the fear that a new protocol might have unforeseen problems three the desire to get the job done before the technology changed in fast ethernet instead of creating a completely new protocol the itle decided to keep all the old packet formats

interfaces and procedural rules and simply reduce the bit time from 100 nond to 10 nanc this effectively increased the bandwidth to 100 Mbps the protocol was officially known as 802.3 U but was more commonly called Fast ethernet the change in bit time presented a number of challenges to the designers of fast ethernet the reduction in bit time increases the number of bits sent within a set time period however these bits still take the same amount of time to travel across a length of wire that is the propagation delay is the same over a given length this means that more bits will be sent during the propagation delay time used as the Collision window either minim minimum frame size need to be increased or the propagation delay need to be reduced by reducing

cable length changing the minimum frame size would have caused problems with a backward compatibility of this standard so the maximum Network size was reduced with typical maximum cable length of 100 m now let us see different media types supported by fast

ethernet fast ethernet defines three basic physical layer specifications as shown in figure 100 base T4 this uses Category 3 four pair twisted pair cable it supports a distance of 100 m and RJ 45 connectors 100 Base TX this uses Category 5 twisted pair which are more expensive than Category 3 cables these are Cat 5 UTP or type 1 STP cable with RJ45 connectors this scheme allows full duplex communication stations can transmit at 100 Mbps and receive at 100

Mbps at the same time 100 base FX this uses two strands of 62.5 or 125 multimode fiber one for each Direction so it is full duplex with 100 Mbps in each Direction in addition the distance between a station and a hub can be up to 2,000 M it supports SC mic or STD connectors there are also few other physical standards in fast ethernet which will be explained now 100 base SX it is a version of fast ethernet over optical fiber it uses two strands of multimode optical fiber for receive and transmit it is a lower cost alternative to using 100 base FX because it uses short wavelength Optics which are significantly less expensive than the long wavelength Optics used in 100 base FX 100 base SX can operate at distances

up to 550 M 100 base BX it is a version of fast ethernet over a single strand of optical fiber single mode fiber is used along with a special multiplexer which splits the signal into transmit and receive wavelengths the two wavelengths used for transmit and receive are 131 nanometer and 1550 nanometer distances supported are 10 20 or 40 kilom 100 base LX 10 it is a version of fast ethernet over two single mode optic fibers it has a nominal reach of 10 kilm and a nominal wavelength of 1310 nanometer now let us move on to layered protocol architecture of fast ethernet various layers of the fast ethernet protocol architecture are shown in figure the MII is medium independent interface between the Mac layer and the physical

layer it allows any physical layer to be used with the Mac layer the MII provides two media status signals one indicates presence of the carrier and the other indicates absence of collision the reconciliation sublayer Maps these signals to physical signaling primitives understood by the physical layer the physical layer is divided into three sublayers physical coding sublayer PCS this sublayer provides a uniform interface to the reconciliation layer for all physical media carrier sense and collision detect indications are generated by this sublayer it also manages the autonegotiation process by which the network interface card communicates with the network to determine the network speed 10 or 100 Mbps and mode of operation whether it is

half duplex or full duplex physical medium attachment PMA this sublayer provides a medium independent means for the PCS to support various serial bit oriented physical media this layer serializes code groups for transmission and deserializes bits received from the medium into code groups physical medium dependent PMD this Maps the physical medium to the PMA it defines the physical layer signaling used for various media so this was about layered protocol architecture of fast ethernet frame format of fast ethernet is same as that of ethernet so friends this was ethernet and fast ethernet finally summarizing we learned about introduction to ethernet which was followed by layered protocol architecture and details of physical and

Mac operations in Ethernet after

ethernet we discussed fast ethernet hope the concepts explained in this lecture were understandable and helpful hope to see you again in the next lecture till then goodbye bye enjoy the day thank you